Calculation Engine

1.0

Generated by Doxygen 1.18.0

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Chapter 1

Calculation Engine

This project demonstrates a small C++ application with a Logger class and a CalculationEngine class. The Logger displays messages, while CalculationEngine performs a power calculation using a base and exponent.

Chapter 2

Todo List

Member CalculationEngine::power (double base, double exponent) const

Add handling for cases where the result may overflow.

Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

CalculationEngine
Performs basic mathematical operations 9
Logger
Writes messages to the standard output stream 11

Chapter 4

File Index

4.1 File List

Here is a list of all files with brief descriptions:

CalculationEngine.h
: Declares a class for basic mathematical calculations 13

Logger.h
: Declares a simple logging utility 14

main.cpp
: Entry point of the calculation engine application 15

Chapter 5

Class Documentation

5.1 CalculationEngine Class Reference

Performs basic mathematical operations.

`#include <CalculationEngine.h>`

Public Member Functions

- CalculationEngine (Logger &log)
  Creates a calculation engine.
- double power (double base, double exponent) const
  Calculates the power of a number.

Private Attributes

- Logger & logger

5.1.1 Detailed Description

Performs basic mathematical operations.

The CalculationEngine class uses a Logger object to report calculation results. It currently provides a power calculation using a base and an exponent.

Definition at line 18 of file CalculationEngine.h.

5.1.2 Constructor & Destructor Documentation

5.1.2.1 CalculationEngine()

```
CalculationEngine::CalculationEngine (
            Logger & log)  [inline], [explicit]
```

Creates a calculation engine.

Parameters

log	A reference to the Logger used by the engine.

Definition at line 27 of file CalculationEngine.h.

```
00028         : logger(log) {}
```

References logger.

5.1.3 Member Function Documentation

5.1.3.1 power()

```
double CalculationEngine::power (
            double base,
            double exponent) const  [inline]
```

Calculates the power of a number.

Parameters

base	The number to be raised to a power.
exponent	The exponent applied to the base.

Returns

The result of base raised to exponent.

Todo Add handling for cases where the result may overflow.

Definition at line 37 of file CalculationEngine.h.

```
00037                                                 {
00038         double result = 1.0;
00039         if (exponent < 0) {
00040             for (int i = 0; i < -exponent; ++i) result *= base;
00041             result = 1.0 / result;
00042         } else {
00043             for (int i = 0; i < exponent; ++i) result *= base;
00044         }
00045         logger.log(LogLevel::INFO, "Power calculation completed.");
00046         return result;
00047     }
```

References INFO, Logger::log(), and logger.

Referenced by main().

5.1.4 Member Data Documentation

5.1.4.1 logger

```
Logger& CalculationEngine::logger  [private]
```

Definition at line 20 of file CalculationEngine.h.

Referenced by CalculationEngine(), and power().

The documentation for this class was generated from the following file:

- CalculationEngine.h

5.2 Logger Class Reference

Writes messages to the standard output stream.

```
#include <Logger.h>
```

Public Member Functions

- Logger (LogLevel level=LogLevel::INFO)

  Creates a Logger with the given minimum log level.
- void log (LogLevel level, const std::string &message) const

  Displays a log message if its level is permitted.

Private Attributes

- LogLevel minimumLevel

5.2.1 Detailed Description

Writes messages to the standard output stream.

The Logger class provides a small interface for displaying messages with a selected logging level.

Definition at line 29 of file Logger.h.

5.2.2 Constructor & Destructor Documentation

5.2.2.1 Logger()

```
Logger::Logger (
            LogLevel level = LogLevel::INFO)  [inline], [explicit]
```

Creates a Logger with the given minimum log level.

Parameters

level	The minimum level of messages to display.

Definition at line 38 of file Logger.h.

```
00039         : minimumLevel(level) {}
```

References INFO, and minimumLevel.

5.2.3 Member Function Documentation

5.2.3.1 log()

```
void Logger::log (
            LogLevel level,
            const std::string & message) const  [inline]
```

Displays a log message if its level is permitted.

Parameters

level	The level of the message.
message	The text to display.

Definition at line 46 of file Logger.h.

```
00046                                                                {
00047         if (static_cast<int>(level) >= static_cast<int>(minimumLevel)) {
00048             std::cout « message « std::endl;
00049         }
00050     }
```

References minimumLevel.

Referenced by CalculationEngine::power().

5.2.4 Member Data Documentation

5.2.4.1 minimumLevel

```
LogLevel Logger::minimumLevel  [private]
```

Definition at line 31 of file Logger.h.

Referenced by log(), and Logger().

The documentation for this class was generated from the following file:

- Logger.h

Chapter 6

File Documentation

6.1 CalculationEngine.h File Reference

Declares a class for basic mathematical calculations.

`#include "Logger.h"`

Classes

- class CalculationEngine

  Performs basic mathematical operations.

6.1.1 Detailed Description

Declares a class for basic mathematical calculations.

Definition in file CalculationEngine.h.

6.2 CalculationEngine.h

Go to the documentation of this file.

```
00001
00005 #ifndef CALCULATION_ENGINE_H
00006 #define CALCULATION_ENGINE_H
00007
00008 #include "Logger.h"
00009
00018 class CalculationEngine {
00019 private:
00020     Logger& logger;
00021
00022 public:
00027     explicit CalculationEngine(Logger& log)
00028         : logger(log) {}
00029
00037     double power(double base, double exponent) const {
00038         double result = 1.0;
00039         if (exponent < 0) {
00040             for (int i = 0; i < -exponent; ++i) result *= base;
00041             result = 1.0 / result;
00042         } else {
00043             for (int i = 0; i < exponent; ++i) result *= base;
00044         }
00045         logger.log(LogLevel::INFO, "Power calculation completed.");
00046         return result;
00047     }
00048 };
00049
00050 #endif
```

6.3 Logger.h File Reference

Declares a simple logging utility.

```
#include <iostream>
#include <string>
```

Classes

- class Logger

  Writes messages to the standard output stream.

Enumerations

- enum class LogLevel { DEBUG , INFO , WARNING , ERROR }

  Defines the available logging levels.

6.3.1 Detailed Description

Declares a simple logging utility.

Definition in file Logger.h.

6.3.2 Enumeration Type Documentation

6.3.2.1 LogLevel

```
enum class LogLevel  [strong]
```

Defines the available logging levels.

Enumerator

DEBUG	Detailed information useful during debugging.
INFO	General information about program execution.
WARNING	Indicates a condition that may need attention.
ERROR	Indicates an error during execution.

Definition at line 15 of file Logger.h.

```
00015                    {
00016     DEBUG,
00017     INFO,
00018     WARNING,
00019     ERROR
00020 };
```

6.4 Logger.h

Go to the documentation of this file.

```
00001
00005 #ifndef LOGGER_H
00006 #define LOGGER_H
00007
00008 #include <iostream>
00009 #include <string>
00010
00015 enum class LogLevel {
00016     DEBUG,
00017     INFO,
00018     WARNING,
00019     ERROR
00020 };
00021
00029 class Logger {
00030 private:
00031     LogLevel minimumLevel;
00032
00033 public:
00038     explicit Logger(LogLevel level = LogLevel::INFO)
00039         : minimumLevel(level) {}
00040
00046     void log(LogLevel level, const std::string& message) const {
00047         if (static_cast<int>(level) >= static_cast<int>(minimumLevel)) {
00048             std::cout << message << std::endl;
00049         }
00050     }
00051 };
00052
00053 #endif
```

6.5 main.cpp File Reference

Entry point of the calculation engine application.

```
#include "Logger.h"
#include "CalculationEngine.h"
#include <iostream>
```

Functions

- int main ()

  Runs the calculation engine example.

6.5.1 Detailed Description

Entry point of the calculation engine application.

Definition in file main.cpp.

6.5.2 Function Documentation

6.5.2.1 main()

```
int main ()
```

Runs the calculation engine example.

Returns

0 when the program finishes successfully.

Definition at line 20 of file main.cpp.

```
00020           {
00021     Logger logger(LogLevel::INFO);
00022     CalculationEngine engine(logger);
00023     double base = 2.0;
00024     double exponent = 3.0;
00025     double result = engine.power(base, exponent);
00026     std::cout « base « " raised to " « exponent
00027               « " = " « result « std::endl;
00028     return 0;
00029 }
```

References INFO, main(), and CalculationEngine::power().

Referenced by main().

6.6 main.cpp

Go to the documentation of this file.

```
00001
00012 #include "Logger.h"
00013 #include "CalculationEngine.h"
00014 #include <iostream>
00015
00020 int main() {
00021     Logger logger(LogLevel::INFO);
00022     CalculationEngine engine(logger);
00023     double base = 2.0;
00024     double exponent = 3.0;
00025     double result = engine.power(base, exponent);
00026     std::cout « base « " raised to " « exponent
00027               « " = " « result « std::endl;
00028     return 0;
00029 }
```

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